

Mass Transfer

Let's begin with some definitions:

ρ_i = mass concentration of species i [kg/m^3]
 C_i = molar concentration of i [kmol/m^3]

$$\rho = \sum \rho_i, \quad m_i = \frac{\rho_i}{\rho}$$

$$C = \sum C_i, \quad x_i = \frac{C_i}{C}$$

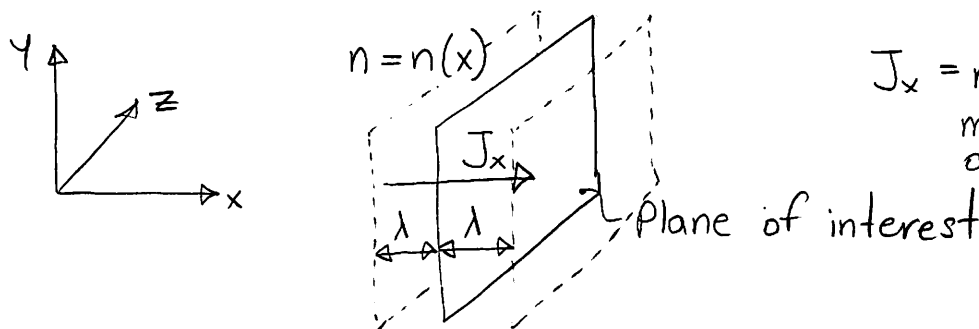
$$C_i = \frac{\rho_i}{M_i}, \quad M = \sum x_i M_i$$

$$C = \frac{\rho}{M}, \quad \frac{1}{M} = \sum \frac{m_i}{M_i}$$

$$\frac{\rho_i}{\rho} = x_i, \quad \rho = \sum \rho_i$$

m_i = mass fraction of i
 ρ = total mass of mixture
 x_i = mole fraction of i
 P_i = partial pressure of i
 P = total pressure

Now let's consider the self-diffusion of a gas.



J_x = mean number of molecules crossing unit area of plane per unit time.

Similar analysis to when we derived viscosity on page 146 of notes.

$$J_x = \frac{1}{6} \bar{v} n_1(x - \lambda) - \frac{1}{6} \bar{v} n_1(x + \lambda)$$

$$= \frac{1}{6} \bar{v} (n_1(x - \lambda) - n_1(x + \lambda)) = \frac{1}{6} \bar{v} \left(-2 \frac{\partial n_1}{\partial x} \lambda \right)$$

$$\boxed{J_x = -D \frac{\partial n_1}{\partial x}} \quad \text{and} \quad \boxed{D = \frac{1}{3} \bar{v} \lambda} \Rightarrow \text{Coefficient of self diffusion. } [\text{m}^2/\text{s}]$$

$\hookrightarrow$ Fick's Law

Note Fick's Law doesn't only apply to self-diffusion. As long as the molecules are similar (ie. N_2 & O_2), it still works.

Other ways of writing it are:

$$j_{m,x} = -D_{i,m} \frac{\partial p_i}{\partial x} = -D_{i,m} p \frac{\partial m_i}{\partial x} \Rightarrow D_{i,m} = \text{diffusion coefficient of } i \text{ in } m.$$

$$\boxed{\vec{j}_{m,i} = -D_{i,m} p \nabla m_i}$$

$$J_{m,x} = -D_{i,m} \frac{\partial C_i}{\partial x} = -D_{i,m} \cdot C \cdot \frac{\partial x_i}{\partial x}$$

$$\boxed{\vec{J}_{m,i} = -D_{i,m} \cdot C \cdot \nabla x_i}$$

For a Lagrangian system:

$$\boxed{\frac{DC_i}{Dt} = D_{i,m} \nabla^2 C_i}$$

Mass Transfer

$$\boxed{\frac{DT}{Dt} = \alpha \nabla^2 T}$$

Heat Transfer

Our governing equation is:

$$\boxed{\frac{DC_i}{Dt} = \frac{\partial C_i}{\partial t} + \vec{v} \cdot \nabla C_i = D_{i,m} \nabla^2 C_i}$$

At steady state and $\vec{v} = 0$

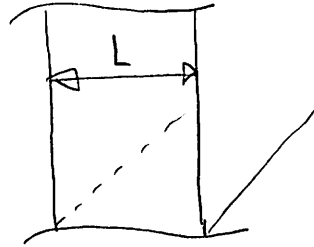
$$\nabla^2 C_i = 0, \quad \nabla^2 x_i = 0, \quad \nabla^2 p_i = 0, \quad \nabla^2 m_i = 0$$

If we have a 1-D problem and steady, with $\vec{v} = 0$

$$N_i = \frac{\Delta C_i}{R_m} \longleftrightarrow \text{Mass transfer resistance}$$

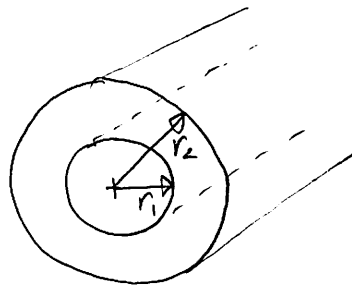
So our mass transfer resistances are:

Slab:



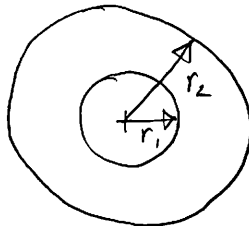
$$R_m = \frac{L}{D_{1,2} A}$$

Cylinder:



$$R_m = \frac{\ln(r_2/r_1)}{2\pi L D_{1,2}}$$

Sphere:



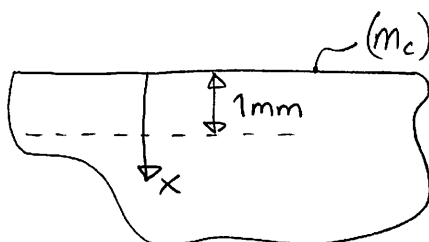
$$R_m = \frac{1}{2\pi D_{1,2}} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Example Case hardening of mild steel.

Mild steel 0.2% or carbon (mass fraction). $m_c = 0.002$

$$\left. \begin{array}{l} t=0, x=0 \quad (m_c)_{x=0} = 0.015 \\ t=?, x=1\text{mm} \quad (m_c)_{x=1\text{mm}} = 0.008 \end{array} \right\} D_{c,s} = 5.6 \times 10^{-10} \text{ m}^2/\text{s}$$

At what time, t , will this happen $(m_c)_{x=1\text{mm}} = 0.008$?
To solve, we first need to model the problem. We can use the semi-infinite body approach.



$$(m_c)_{x=0, t=0} = 0.015$$

$$\frac{(m_c)_{x=1\text{mm}} - (m_c)_\infty}{(m_c)_{x=0\text{mm}} - (m_c)_\infty} = \text{erfc} \left(\frac{x}{2\sqrt{D_{c,s}t}} \right)$$

$$\frac{6}{13} = 0.4615 = \text{erfc}(\beta) \Rightarrow \text{Look in Mills pg. 924}$$

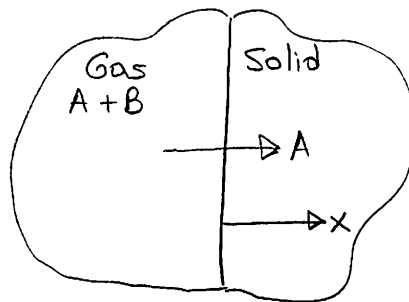
$$\beta = 0.52$$

So our solution becomes:

$$0.52 = \frac{10^{-3} \text{ m}}{2\sqrt{Dt}} \Rightarrow \boxed{t = 1650 \text{ s} = 28 \text{ minutes}}$$

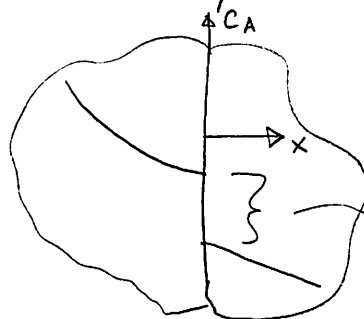
Where heat and mass transfer begin to differ is the fact that mass transfer allows for equilibrium to be reached with a concentration difference.

For example:



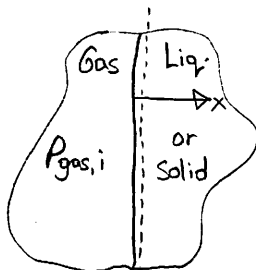
$$-CD_{AB} \frac{\partial x_A}{\partial x} \Big|_{x=0} = J_{A,s} \Big|_{x=0}$$

So our concentration may look like:



Discontinuity (Can't happen with temperature!)

To deal with this, we can use Henry's Law to solve for mole fraction just inside:



~ Just inside the Liquid or Solid.

$$\boxed{P_{\text{gas},i} = X_{i,u} \cdot H} ; H = \text{Henry's constant}$$

H (1 atm (10^5 Pa)) $\Rightarrow$ in Water

	290 K	300 K	320 K	330 K	340 K
O ₂	38 000	45 000	52 000	57 000	65 000
CO ₂	1 280	1 710	2 170	3 220	4 500
Air	62 000	74 000	92 000	99 000	(224)

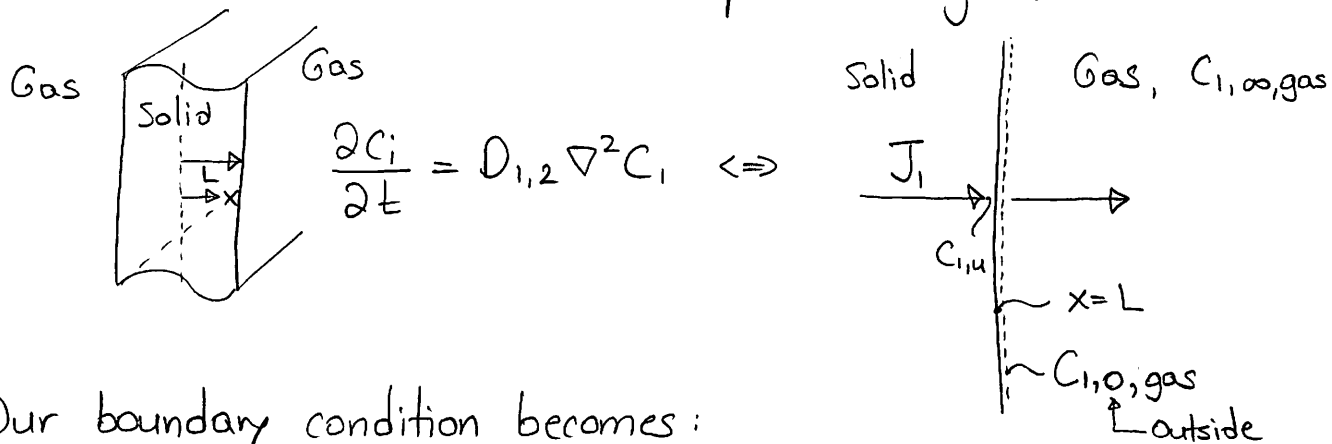
If we treat the gas and solid as a solution, we can say:

$$\boxed{C_{i,u} = S P_{\text{gas},i}} \quad , \quad P_{\text{gas},i} = \text{partial pressure of gas } i \text{ next to the interface}$$

$$C_{i,u} = \text{kilomoles of gas } i \text{ per cubic meter of solid} \left[\frac{\text{kmol}}{\text{m}^3} \right]$$

$$S = \text{solubility}$$

So now if we look at a slab problem again:



Our boundary condition becomes:

$$\underbrace{-D_{1,2} \left(\frac{\partial C_1}{\partial x} \right) \bigg|_{x=L}}_{\text{Inside the solid}} = J_1 = \underbrace{h_m (C_{1,0,gas} - C_{1,\infty,gas})}_{\text{Outside, in the gas}} \quad (1)$$

$h_m = \text{mass transfer coeff.} \left[\frac{\text{m}}{\text{s}} \right]$

Now we can solve our problem.

Multiplying (1) by S/S ($C_{1,u} = C_{1,gas} \cdot S$)

$$-D_{1,2} \left(\frac{\partial C_1}{\partial x} \right) = \frac{h_m}{S} [(C_{1,0} - C_{1,\infty})_{\text{gas}} \cdot S] \quad (2)$$

Non-dimensionalizing equation (2)

$$\bar{x} = \frac{x}{L} \quad , \quad \phi = \frac{C_1 - C_{1,u,\infty}}{C_{1,u,0} - C_{1,u,\infty}}$$

$$d\bar{x} = \frac{dx}{L} \quad , \quad d\phi = \frac{dC_1}{C_{1,u,0} - C_{1,u,\infty}}$$

$$C_1 = C(x=L, t) \text{ inside solid}$$

$$C_{1,u,\infty} = C(t \rightarrow \infty) \text{ inside}$$

$$C_{1,u,0} = C(t \rightarrow 0) \text{ inside}$$

$$C_{1,0,gas} = C \text{ at interface outside in gas}$$

$$C_{1,\infty,gas} = C(x \rightarrow \infty) \text{ outside}$$

$$-D_{1,2} \frac{1}{L} \frac{\partial \phi}{\partial \bar{x}} (C_{1,u,0} - C_{1,u,\infty}) = \frac{h_m}{S} [(C_{1,0} - C_{1,\infty})_{\text{gas}} \cdot S]$$

But note that $C_{1,0,\text{gas}} \cdot S = C_1$ and $C_{1,\infty,\text{gas}} \cdot S = C_{1,u,\infty}$

$$-D_{1,2} \frac{1}{L} \frac{\partial \phi}{\partial \bar{x}} (C_{1,u,0} - C_{1,u,\infty}) = \frac{h_m}{S} (C_1 - C_{1,u,\infty})$$

$$-D_{1,2} \frac{1}{L} \frac{\partial \phi}{\partial \bar{x}} = \frac{h_m}{S} \underbrace{\left(\frac{C_1 - C_{1,u,\infty}}{C_{1,u,0} - C_{1,u,\infty}} \right)}_{\phi}$$

$$-D_{1,2} \frac{1}{L} \frac{\partial \phi}{\partial \bar{x}} = \frac{h_m \phi}{S}$$

$$\boxed{-\frac{\partial \phi}{\partial \bar{x}} = \left(\frac{h_m L}{D_{1,2} S} \right) \phi} \Rightarrow \text{Similar to heat transfer solution. Just added factor } S.$$

$Bi_m = \text{mass transfer Biot number}$
 $= \frac{\text{internal diffusion resistance}}{\text{external convection resistance}}$

If we look back at our heat transfer analogy

